

PRADEEP PANDEY

Target Profile · November 2028

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RESEARCH INTERESTS

Robustness and interpretability of aligned behavior in open-weight language models: how refusal and other safety behaviors hold up under perturbation, what activation-level structure underlies them, and how to evaluate both rigorously at low compute.

EDUCATION

Tribhuvan University: Bachelor of Computer Application (BCA) · Kathmandu, Nepal · Expected Nov 2028

- Final-year project: an activation-probing evaluation suite for small open-weight LLMs, co-supervised with TU faculty as a companion study to the preprint below.
- Additional training: ARENA Chapters 0, 1, 3 (self-directed, 2027–2028); NPTEL (IIT) certified Calculus, credit-bearing (2026).

PUBLICATIONS AND PREPRINTS

- [1] Pandey, P. (2028). *How Stable Is Refusal? Behavioral and Activation-Level Robustness in Small Open-Weight Language Models*. arXiv:2808.04217 [cs.LG]. Accepted at **BlackboxNLP 2028** (co-located with EMNLP 2028). Code and data: github.com/pradeepandey-r/refusal-stability.
- [2] Pandey, P. (2027). *Independent Replication of “Refusal in Language Models Is Mediated by a Single Direction” (Arditi et al., 2024), extended to Gemma-2-2B and Qwen2.5-1.5B*. Technical note. Code: github.com/pradeepandey-r/refusal-direction-replication.

RESEARCH EXPERIENCE

Independent Researcher: Refusal-Robustness Study · Kathmandu, Nepal · Mar 2028 – Aug 2028

- Designed and pre-registered a robustness study of refusal behavior: 240 refusal-eliciting base prompts from public safety benchmarks under 5 semantics-preserving perturbation families, run against six open-weight models (1.5B–9B); 8,640 completions scored on an automated Inspect harness for \$287 of a \$500 compute envelope.
- Core finding of [1]: worst-case behavioral refusal consistency fell by 31 percentage points under multi-turn decomposition, while linear probes on residual-stream activations (TransformerLens) retained AUC at or above 0.92; behavioral robustness and internal detectability come apart under perturbation.

SPAR (Supervised Program for Alignment Research): Research Fellow · Remote · Sep 2027 – Dec 2027

- Selected for the Fall 2027 cohort. Investigated scale-sensitivity of refusal-direction ablation across three open-weight model families under the mentorship of an alignment researcher; the cohort project's experimental design became the seed of [1].

NAAMII: Research Intern, Applied ML · Lalitpur, Nepal · Mar 2028 – May 2028

- Built the group's internal robustness-evaluation pipeline for open-weight LLMs on Inspect (14 task suites, versioned datasets, cached generations), replacing ad-hoc per-project scripts; the pipeline is reused across two ongoing projects at the institute.

INDUSTRY EXPERIENCE

Flipkart: Machine Learning Intern, Trust and Safety · Remote (Bengaluru, India) · Jun 2027 – Aug 2027

- Shipped an offline regression-evaluation suite for a production abuse-text classifier: frozen evaluation sets, drift checks against weekly traffic samples, and a release scorecard the team adopted for model sign-off.

OPEN-SOURCE CONTRIBUTIONS

TransformerLens: Contributor (feature merged) · Remote · Jan 2028

- Wrote, tested, and merged a batched activation-caching utility for the hooks API, with documentation and CI tests; the same code path drives the analysis pipelines behind [1] and [2].

SELECTED WRITING AND GRANTS

Nine technical articles at pradeeppandey.name.np and six LessWrong / Alignment Forum posts on evaluation methodology and interpretability, including *Designing a Small-Scale Safety Experiment* (2028) and *Anatomy of the Transformer* (2027). Manifund micro-grant: independent compute funding for the refusal-robustness study (2027).

TECHNICAL SKILLS

Python, PyTorch, TransformerLens, Inspect (UK AISI), lm-evaluation-harness, Hugging Face Transformers, NumPy, pandas, Git, Linux, LaTeX